Arithmetic formulas for low-product tanh networks and a Bayesian learning converse

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Abstract

We prove that every Boolean function with a globally unit-margin tanh representation of Frobenius product P has a logarithmic-depth integer arithmetic circuit of size polynomial in $d+P$ whose output sign equals the function. The original network may have arbitrary finite depth, widths, and real coefficients. In particular, polynomial product implies nonuniform PNC^1, hence logarithmic space with polynomial advice. The analytic ingredient is an exact Fourier–Poisson estimator for tanh that multiplies expected arithmetic work times second moment by only a universal constant. Cost-aware neuron sampling makes this potential propagate according to squared Frobenius norms. A logarithmic-depth tanh normal form, finite-set derandomization, coefficient rounding, and real-affine formula balancing complete the proof.

For the original Gaussian Bayesian network conditioned on a global absolute output margin and exact training labels, an established posterior representation ceiling then gives a converse: a distribution-free guarantee after n samples, with confidence-failure mass below any fixed constant less than one half, forces a small arithmetic-sign representation of the target. This applies to the strong-prior limit and every fixed Gaussian exponent $\gamma > 1$ at the specified sufficiently large widths. It handles the full posterior mixture. The endpoint $\gamma = 1$ is not covered by this strengthening. No unconditional exponential-depth sample lower bound or general polynomial-depth learning upper bound follows. The new arguments have undergone mathematical review but have not been formalized in Lean.

1 Introduction and results

How much sequential computation can a neural network hide behind a small product of layer norms? This question matters for strong Gaussian Bayesian priors: after conditioning on data, most posterior functions can have low-product representatives even when the sampled parameters themselves are numerous and not small. A structural theorem about these functions can therefore turn a learning guarantee into a necessary computational condition on the target.

For tanh networks, we show that polynomial Frobenius product restricts the represented Boolean function to a class between Boolean logarithmic depth and logarithmic space. This conclusion is stronger than merely exhibiting a small Boolean circuit. It also differs from simulating the given network layer by layer: the original network can have unbounded finite depth and arbitrary real weights.

The proof has two parts. A geometric covering argument reduces the function to an ordinary-tanh network of logarithmic depth and polynomial resources. We include this normal form and its required width compression in the appendices. We then replace the remaining nonlinear computation

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by a small collection of randomized arithmetic expressions. An exact Fourier representation and Poisson random products prevent second-moment losses from escalating with the moment order. The useful quantity is the product of expected work and second moment. It grows by a constant times the squared layer Frobenius norm, so logarithmic retained depth leaves polynomial cost. The probabilistic method selects expressions correct on the whole Boolean cube. A careful finite-bit conversion makes their coefficients legitimate circuit constants.

1.1 Conventions

Write $X_d = \{-1, 1\}^d$ and augment the input by one constant coordinate. A depth-D network has D weighted layers,

$$h^0(x) = (1, x), \qquad h^s(x) = \tanh(W^s h^{s-1}(x)) \ (s < D), \qquad z(x) = W^D h^{D-1}(x).$$

The readout is scalar and linear. Hidden widths are arbitrary finite integers. There are no later free biases; a constant must be carried by ordinary hidden neurons. Put

$$P_F(W) = \prod_{s=1}^{D} \|W^s\|_F .$$

A representation of $f : X_d \to \{-1, 1\}$ has *global unit margin* if $f(x)z(x) \geq 1$ for every $x \in X_d$. Purely affine representations ($D = 1$) are allowed.

An integer arithmetic circuit uses binary $+$ and multiplication gates, input bits, and constants $-1, 0, 1$. The nonuniform class PNC^1 consists of signs of polynomial-size, logarithmic-depth such circuits: the accepting output is strictly positive. This is an arithmetic sign class, not Boolean NC^1. Standard results give

$$\mathsf{NC}^1 \subseteq \mathsf{PNC}^1 \subseteq \mathsf{L/poly} \quad \text{in the nonuniform setting,} \qquad \mathsf{AC}^0(\mathsf{PNC}^1) = \mathsf{PNC}^1.$$

Here and below the notation PNC^1 is explicitly nonuniform unless a uniform evaluator is mentioned. The last display uses the corresponding nonuniform circuit form of the standard closure theorem [1]; the advice consists of the selected finite circuits.

Theorem 1.1 (Low product gives an arithmetic-sign circuit)**.** *There are universal constants $a, c > 0$ such that any $f : X_d \to \{-1, 1\}$ with a global unit-margin tanh representation of product at most P has an integer arithmetic circuit of size at most $(d + P + 2)^a$ and depth at most $c \log(d + P + 2)$ whose output is positive exactly when $f(x) = 1$. Its only constants are $-1, 0, 1$. Consequently, any family with $P = \mathrm{poly}(d)$ belongs to nonuniform $\mathsf{PNC}^1 \subseteq \mathsf{L/poly}$.*

The theorem is existential. It does not say that the original real parameters are computable, that the circuit can be found efficiently, or that the original network can be evaluated in logarithmic space from an arbitrary real encoding. It gives another finite representation of the same Boolean function.

Theorem 1.2 (Bayesian consequence, informal)**.** *For the Gaussian posterior specified in Section 6, fix either an exponent $\gamma > 1$ and a sufficiently large finite learner width, or a sufficiently large fixed width followed by the strong-prior limit. Suppose that for every input distribution, n labeled samples have positive probability of giving posterior probability greater than $2/3$ to the correct label outside a set of mass at most $\varepsilon < 1/2$. Then the target has an arithmetic-sign circuit of size polynomial in $d + n + (1 - 2\varepsilon)^{-1}$ and logarithmic depth in the same quantities. The polynomial is independent of learner depth and width. Precise width and limit quantifiers appear in Theorems 6.1 and 6.2.*

If a single distribution-free sample bound polynomial jointly in the resources of arbitrary bounded-weight deep tanh teachers and learner depth held in either of these regimes, it would follow that P/poly is contained in nonuniform PNC^1, and hence $\mathsf{P}/\mathrm{poly} = \mathsf{L}/\mathrm{poly}$; see Corollary 6.4. This is a conditional obstruction, not a proved separation. The question whether unrestricted tanh Bayesian learning has polynomial or necessarily exponential depth dependence remains open.

1.2 Relation to arithmetic neural computation

Merrill, Jiang, Li, Lin, and Sabharwal relate linear recurrent networks to logarithmic-depth arithmetic circuits and distinguish them from nonlinear recurrent architectures [3]. Our setting is a different restriction: the network is genuinely nonlinear and feedforward, and neither its original depth nor its width is bounded. The hypothesis is low Frobenius product together with a global Boolean margin. The Fourier–Poisson argument constructs a new arithmetic representation from this norm constraint; it does not replace the nonlinear recurrence by an exact linear recurrence. The cited recurrent results therefore do not supply the theorem here. The arithmetic-class closure and logspace evaluation facts themselves are established results of counting complexity [1, 2].

2 An exact random arithmetic expression for tanh

Expressions in this section may use complex arithmetic internally and real or imaginary part at a unary gate. Their final outputs are real. Cost counts leaves and arithmetic operations. Every child call uses independent fresh randomness, and the random expression's shape is independent of its input. Sampling exact real constants is a nonuniform representation device; no computational cost bound for an exact sampling algorithm is asserted.

Lemma 2.1 (Fourier identity)**.** *For real u,*

$$\tanh u = \int_0^\infty k(\omega)\sin(\omega u)\,d\omega, \qquad k(\omega) = \frac{1}{\sinh(\pi\omega/2)}, \qquad \int_0^\infty \omega k(\omega)\,d\omega = 1.$$

The first integral is absolutely convergent.

Proof. The expansion $k(\omega) = 2\sum_{j\ge 0} e^{-(j+1/2)\pi\omega}$, combined with integration of sine against an exponential, gives

$$\int_0^\infty k(\omega)\sin(\omega u)\,d\omega = 2\sum_{j\ge 0}\frac{u}{(j+1/2)^2\pi^2 + u^2} = \tanh u.$$

The last equality is the partial-fraction identity obtained by logarithmically differentiating the Euler product

$$\cosh u = \prod_{j\ge 0}\left(1 + \frac{u^2}{(j+1/2)^2\pi^2}\right).$$

Termwise integration is justified by $\int k(\omega)|\sin(\omega u)|\,d\omega < \infty$: near zero $k(\omega) = O(1/\omega)$, and at infinity it decays exponentially. The second integral equals $2\sum_{j\ge 0}((j+1/2)\pi)^{-2} = 1$. □

Theorem 2.2 (A universal work–second-moment transform)**.** *Suppose random real arithmetic expressions $X(x)$ obey, uniformly on a finite input set,*

$$\mathbb{E}X(x) = \mu(x), \quad \mathbb{E}X(x)^2 \le V > 0, \quad \mathbb{E}\,\mathrm{cost}(X) \le T, \quad T \ge 1.$$

There is a random arithmetic expression $Y(x)$ with mean $\tanh\mu(x)$ and bounds $V', T' \geq 0$, $T' \geq 1$, satisfying

$$\mathbb{E}Y(x)^2 \leq V', \qquad \mathbb{E}\,\mathrm{cost}(Y) \leq T', \qquad V'T' \leq CVT$$

for a universal constant C. The case $V = 0$ is handled by the zero expression. For every $s > 0$, the same potential bound holds for the scaled activation $\tanh(s\mu)/s$.

Proof. For fixed $\omega > 0$, put $\lambda = 1 + \omega^2 V$, take $K \sim \mathrm{Poisson}(\lambda)$ independently, and define

$$Z = \prod_{j=1}^{K} (1 + i\omega X_j/\lambda), \qquad S = \mathrm{Im}\, Z.$$

The empty product is one. The Poisson generating function and fresh independence give the exact identities

$$\mathbb{E}Z = e^{i\omega\mu}, \qquad \mathbb{E}|Z|^2 = \exp\left(\frac{\omega^2 \mathbb{E}X^2}{\lambda}\right) \leq e.$$

These require only second moments. Since $\mu^2 \leq V$,

$$\begin{aligned}\mathbb{E}S^2 &\leq \mathbb{E}|Z-1|^2 \\ &= \exp\left(\frac{\omega^2\mathbb{E}X^2}{\lambda}\right) - 2\cos(\omega\mu) + 1 \leq C\omega^2 V.\end{aligned}$$

Together with the preceding bound this proves $\mathbb{E}S^2 \leq C\min(1, \omega^2 V)$. The expected work conditional on ω is at most $CT\lambda$.

Define

$$\begin{aligned} b(\omega) &= \min(1, \omega\sqrt{V}), \\ Z_V &= \int_0^\infty k(\omega) b(\omega)/\sqrt{\lambda}\, d\omega, \\ A_V &= \int_0^\infty k(\omega) b(\omega)\sqrt{\lambda}\, d\omega. \end{aligned}$$

Both integrals are finite and positive. Sample ω with density $q_V(\omega) = k(\omega)b(\omega)/(\sqrt{\lambda} Z_V)$ and output $Y = k(\omega)S/q_V(\omega)$. The bound on $\mathbb{E}|S|$ gives absolute integrability, so Fubini and Lemma 2.1 prove $\mathbb{E}Y = \tanh\mu$. Moreover

$$\mathbb{E}Y^2 \leq C Z_V A_V, \qquad \mathbb{E}\,\mathrm{cost}(Y) \leq CT A_V/Z_V.$$

For $r \geq 0$, $r \leq \min(1, r)\sqrt{1 + r^2} \leq \sqrt{2}r$, hence $\sqrt{V} \leq A_V \leq \sqrt{2V}$. Multiplication proves the claim. Since $A_V/Z_V \geq 1$, fixed overhead is absorbed. Apply this construction to sX and divide its result by s to obtain the last assertion; the s^2 factors cancel in the potential. □

3 Cost-aware sampling through a network

Lemma 3.1 (Linear combination)**.** *Suppose child expressions have means μ_j, second-moment bounds $V_j > 0$, and expected-work bounds $T_j \geq 1$. A linear combination $\sum_j w_j\mu_j$ has an unbiased expression with potential at most*

$$C\left(\sum_j |w_j|\sqrt{V_j T_j}\right)^2.$$

Proof. Omit zero terms. Set

$$L_0 = \sum_j |w_j|\sqrt{V_j/T_j}, \qquad M_0 = \sum_j |w_j|\sqrt{V_j T_j}, \qquad p_j = |w_j|\sqrt{V_j/T_j}/L_0.$$

Sample J with probabilities p_j and output $w_J X_J/p_J$. Its mean is the required sum, its second moment is at most $L_0 M_0$, and expected work is at most CM_0/L_0. Here $M_0/L_0 \geq 1$, which absorbs multiplication overhead. Their product is as claimed. If all terms vanish, use the zero expression. □

Proposition 3.2 (Frobenius propagation)**.** *A tanh network with h hidden layers, input coordinates of absolute value at most one, and product P has an unbiased scalar-output expression whose work–second-moment potential is at most*

$$C^{h+1}(d+1)P^2.$$

For a vector readout, the sum of its coordinate potentials has the same bound. The conclusion also allows neuron-specific normalized tanh activations or identity activations.

Proof. Input coordinates have $V = T = 1$. If $K_j = V_j T_j$ denotes a child potential, Lemma 3.1 and Theorem 2.2 give

$$K_i \leq C\left(\sum_j |w_{ij}|\sqrt{K_j}\right)^2, \qquad \sum_i K_i \leq C\,\|W\|_F^2 \sum_j K_j.$$

The second inequality is rowwise Cauchy–Schwarz. Iterate it through the hidden layers and final readout. All multiplication subexpressions are fresh independent calls; no higher moments are needed. □

The factor C^{h+1} is not a depth-independent constant. The next normal form is what makes the final estimate polynomial.

Theorem 3.3 (Short ordinary-tanh normal form)**.** *There are universal constants a, c such that a globally unit-margin tanh Boolean representation of product P has another globally unit-margin ordinary-tanh representation with at most $c\log(P+2)$ weighted layers, width, individual weights, and product at most $(P+2)^a$, using only its at most P^2 relevant coordinates. The weights can be dyadic with at most $c\log^2(P+2)$ bits per entry after changing the constants.*

Appendices A–B give a self-contained proof. In particular this statement does not replace tanh by arbitrary lookup activations. An affine final normal form is allowed and causes no difficulty in Proposition 3.2.

4 Selecting finite formulas and bounding their coefficients

Apply Theorem 3.3. Write $B = d + P + 2$ and increase a universal exponent a as necessary. Its depth is at most $a\log B$; its dimensions, product, and physical coefficient magnitudes are at most B^a. Its coefficient bit lengths are at most B^a. In particular every nonzero coefficient is at least $\exp(-B^a)$ in absolute value. The deliberately loose exponential bound is sufficient.

The output estimator Y from Proposition 3.2 has deterministic bounds V, T with

$$VT \leq B^a. \tag{1}$$

The margin implies $V \geq 1$, so both final-output bounds are polynomial. Internal bounds need not individually be polynomial. We next bound their logarithms, rather than assuming that all internal scales are well conditioned.

Lemma 4.1 (Internal constants)**.** *All nonzero internal deterministic bounds V, T, L_0, M_0, Z_V, and their reciprocals, lie between $\exp(-B^b)$ and $\exp(B^b)$ for a universal b.*

Proof. The explicit estimator bounds may be used as the recursive definitions of V and T, with all work bounds at least one. If preceding second-moment and work bounds have logarithmic magnitudes at most R, the formulas for L_0, M_0 show that their logarithmic magnitudes are at most $C(R + B^a)$. A nonzero row has a nonzero coefficient at least $\exp(-B^a)$ and at least one retained child. Zero children or zero rows are omitted. The same estimate applies to $V_{\text{in}} = L_0 M_0$ and $T_{\text{in}} = C M_0 / L_0$.

The transform uses

$$\sqrt{V} \le A_V \le \sqrt{2V}, \qquad c \min(\sqrt{V}, 1) \le Z_V \le C \min(\sqrt{V}, 1). \tag{2}$$

For $V \le 1$, integration over $[1, 2]$ gives the lower bound for Z_V, and $b(\omega) \le \omega\sqrt{V}$ gives the upper bound. For $V \ge 1$, integrate over $[0, 1/\sqrt{V}]$ for the lower bound, using $k(\omega) \asymp 1/\omega$ there. For the upper bound split at $1/\sqrt{V}$; the remaining integral is at most $\int_{1/\sqrt{V}}^{\infty} k(\omega)/(\omega\sqrt{V})\, d\omega$, which is uniformly bounded by a further split at one. Thus $V' = C Z_V A_V$ and $T' = C T A_V / Z_V$ satisfy a logarithmic recurrence $R_{s+1} \le C R_s + C B^a$. There are $O(\log B)$ layers, so its solution is $B^{O(1)}$. □

Lemma 4.2 (A good realization of polynomial size)**.** *There is a fixed collection of at most $C(d+1)$ arithmetic formulas of total size polynomial in B such that, on every input, more than two thirds of their outputs have the correct sign and absolute margin at least $1/2$. Every coefficient in these formulas has modulus at most $\exp(B^c)$ for a universal c.*

Proof. Average $n_0 = \lceil 40V \rceil$ independent output estimators in one group. Chebyshev gives error at most $1/2$ with probability at least 0.9 at each input. Form $r = O(d+1)$ independent groups, with a sufficiently large universal constant. Chernoff and a union bound over 2^d inputs give probability at least 0.9 that more than two thirds of the groups are good at every input. Their expected total size is at most $S = B^{O(1)}$, by (1) and $V \ge 1$. Markov gives realized size at most $20S$ except with probability at most $1/20$.

Uniformly over all internal neuron types, Lemma 4.1 and the density formula imply

$$\mathbb{P}(\omega < \eta) \le e^{B^b} \eta, \qquad \mathbb{P}(\omega > U) \le e^{B^b} e^{-c_0 U} \quad (U \ge 1).$$

The first estimate uses $k(\omega) \le C/\omega$ and $b(\omega) \le \omega\sqrt{V}$; the second uses exponential decay and the lower bound on Z_V. A newly reached node draws its fresh frequency after its existence and type have been determined. Summing the conditional bad-frequency probabilities over all tree addresses therefore bounds the probability of any bad frequency by their common upper bound times the expected number of reached nodes, at most S. Choose $\eta = e^{-B^{c_1}}$ and $U = B^{c_1}$ with a sufficiently large universal c_1. This failure probability is at most $1/20$.

All three good events have positive joint probability. Fix such a realization. Every Poisson count is at most its total size. The arithmetic constants are obtained from original constants and

$$\frac{w_j}{p_j} = \operatorname{sign}(w_j) L_0 \sqrt{T_j / V_j}, \quad \frac{\omega}{\lambda}, \quad \frac{k(\omega)}{q_V(\omega)} = \frac{Z_V \sqrt{\lambda}}{\min(1, \omega\sqrt{V})}, \quad n_0^{-1}.$$

Their logarithmic magnitudes are polynomial in B on the good event. Each good group approximates an output of absolute value at least one to error at most $1/2$, giving the margin. We have selected an event from the original unbiased construction; we have not conditioned the estimator on truncated frequencies and then asserted unbiasedness. □

Lemma 4.3 (Dyadic rounding and integer numerators)**.** *The formulas in Lemma 4.2 can be replaced by polynomial-size formulas over Gaussian integer constants, addition, multiplication, and real/imaginary-part gates, with all integer constants of polynomial bit length, so that their majority of signs still equals f on the whole cube.*

Proof. Regard each occurrence of a constant in a formula as a separate pair of real variables. A formula of size s is separately linear in each occurrence, with degree at most s and at most exponentially many terms. Real/imaginary-part gates are real-linear. On a coefficient box of magnitude $e^{B^{O(1)}}$, values and first coefficient derivatives are therefore at most $e^{B^{O(1)}}$ because $s = B^{O(1)}$. Equivalently this follows by direct induction through the gates. Round every coefficient component to dyadic error 2^{-B^c} with a sufficiently large constant exponent. Every group output changes by less than $1/8$, uniformly on the cube. Its previously good outputs retain the correct sign and margin at least $3/8$. Bad groups may change sign, but more than two thirds remain good at every input.

Clear denominators with positive integer subtree denominators. At an addition replace $N_1/D_1 + N_2/D_2$ by $(N_1D_2 + N_2D_1)/(D_1D_2)$; at multiplication multiply numerators and denominators. Real and imaginary part commute with division by a positive real denominator. Each denominator is a precomputed constant whose bit length is at most the sum of leaf denominator bit lengths in the subtree. Each child numerator occurs only once, so the result is a polynomial-size formula, not an exponentially expanded circuit. The sign of its real output is that of its integer numerator. □

5 Balancing and the arithmetic complexity class

Lemma 5.1 (Real-affine balancing of complex formulas)**.** *A size-s formula over Gaussian integer constants, complex addition and multiplication, and unary real/imaginary-part gates has an equivalent real integer formula of polynomial size and depth $O(\log(s+2))$. Its constants have no larger bit lengths, up to a constant, than the original constants.*

Proof. First balance the complex formula. Select a subtree Q whose size lies between one third and two thirds of the full size, up to a fixed additive constant. Such a subtree exists for a tree of fan-in at most two, including unary gates. Since Q occurs only once, its surrounding context is real-affine. Thus

$$F = (F[Q := 1] - F[Q := 0])\,\mathrm{Re}\,Q + (F[Q := i] - F[Q := 0])\,\mathrm{Im}\,Q + F[Q := 0].$$

Both Q and each substituted context have at most two thirds of the original size, up to fixed additive terms. Recursing introduces only a fixed number of copies at each recursive level. The resulting size is polynomial in s and depth is $O(\log(s+2))$. The only new constants are $0, 1, i, -1$.

Now replace each complex gate by its fixed-size real/imaginary pair of integer arithmetic gates. Expansion into formulas duplicates subexpressions by only a fixed factor per level. Since the balanced depth is logarithmic, this gives only a polynomial total factor. Doing this expansion before balancing would not justify the bound. The final output is real, so keep its real component. □

Proof of Theorem 1.1. Lemmas 4.2–5.1 give polynomially many polynomial-size logarithmic-depth integer formulas. Their polynomial-bit constants can be built using $-1, 0, 1$: construct a power 2^j as a balanced product of j copies of $1 + 1$, and express an integer as a balanced sum of its signed nonzero powers of two. This adds polynomial size and logarithmic depth. Replace sign inputs by $2b - 1$ for bit inputs. Each formula's positivity predicate belongs to nonuniform PNC^1.

The final majority is itself a PNC^1 predicate (indeed it is the sign of an affine integer sum of the input bits). It is applied to polynomially many PNC^1 predicates. The composition theorem

$[\mathsf{PNC}^1] \cdot [\mathsf{PNC}^1] = \mathsf{PNC}^1$ in [1, Theorem 4.2] therefore gives the claimed circuit with universal polynomial size and logarithmic depth. Equivalently use the constant nesting depth of positivity gates and [1, Proposition 3.4 and Theorem 4.1]. The inclusion in $\mathsf{L/poly}$ follows from their Proposition 2.4, with the selected circuit supplied as advice. Formula balancing alone would not establish Boolean NC^1. If the final integer output can be zero, replacing it by twice that output minus one gives strictly positive and strictly negative values on the two label classes without changing the resource bounds. □

6 A converse for the original Gaussian posterior

Fix $d, n \geq 1$, learner weighted depth $L \geq 3$, and common hidden width N. Write σ^2 for the common variance in energy coordinates. The independent physical weight variances are

$$W_{ij}^1 : \frac{N\sigma^2}{d}, \qquad W_{ij}^s : \sigma^2 \ (2 \leq s < L), \qquad v_i : \frac{\sigma^2}{N}.$$

Equivalently the Gaussian energy is

$$\kappa_N(W) = \frac{d}{N} \left\| W^1 \right\|_F^2 + \sum_{s=2}^{L-1} \|W^s\|_F^2 + N \|v\|_2^2 .$$

For a consistently labeled sample $S = ((x_i, y_i))$, the posterior $Q_{S,\sigma}$ conditions the raw prior on

$$|z(x)| > 1/2 \quad \text{for every } x \in X_d, \qquad y_i z(x_i) > 0 \quad \text{for every training example.}$$

No hidden robustness, internal margin, or parameter bound is imposed. Its induced distribution on Boolean truth tables will also be denoted by $Q_{S,\sigma}$. Define $p_{S,\sigma}(f(x) \mid x) = Q_{S,\sigma}\{h : h(x) = f(x)\}$. The exponent convention is $\sigma^2 = N^{-\gamma}$.

6.1 The precise imported posterior input

Let $\mathcal{H}_P$ be the finite set of Boolean functions on X_d admitting some globally unit-margin tanh representation of product at most P, with arbitrary finite depth and widths. The following posterior theorem is an input from the existing Gaussian analysis [4]; it is not a consequence of the Fourier estimator. We state all its quantitative data so its use here is unambiguous.

Theorem 6.1 (Finite-scale representation ceiling, imported)**.** *Fix $\zeta > 0$ and $0 < \beta < 1$. Define*

$$\begin{aligned}
C_0 &= n + 1, \qquad R_0 = \max\{4(n+1), (d+1)\log(16(n+1))\}, \\
M &= \left\lceil \max\{4, C_0, 2R_0, dC_0(d+1)R_0^2\} \right\rceil, \qquad T_0 = MC_0R_0^2, \\
K_0 &= 2 + (L+14)(2T_0)^{1/(L-1)}, \\
b_0^2 &= \max\{2(d+1)\log 2 + 4\log 8,\ 16\log 9 + 8\log(8L)\}, \\
A_0 &= \left(\frac{K_0+2}{L-1}\right)^{(L-1)/2}, \\
\xi &= \min\left\{1, \frac{1}{4\sqrt{L(K_0+1)}}, \frac{1}{8\sqrt{2}LA_0}\right\}, \\
N_0 &= \left\lceil \max\{M, 2, (b_0/\xi)^{2/\zeta}, 2\log(2/\beta)\} \right\rceil, \\
P_0^2 &= d^{-1}e^{19}(2T_0)^{3/2}.
\end{aligned}$$

For every $N \geq N_0$, every actual positive scale $\sigma^2 \leq N^{-(1+\zeta)}$, and every consistently labeled sample of at most n inputs,

$$Q_{S,\sigma}(\mathcal{H}_{P_0}) \geq 1 - \beta.$$

The threshold N_0 is polynomial in $d, n, L, \log(2/\beta)$ with degree depending only on ζ. The product bound satisfies

$$P_0 \leq e^{19/2} 544^{3/4} (d + n + 2)^9,$$

independently of L, N, ζ, β.

The low-product representative can differ from the sampled parameter vector. The theorem does not assert that posterior weights themselves have bounded energy. Its proof uses a uniform memorizing witness, a Gaussian perturbation set, and Gaussian distance concentration. The exact source is the theorem labeled `thm:posterior-energy-ceiling` in the file identified in [4]; the package dependency manifest records its hash. For the final displayed polynomial, put $S_0 = d+n+2$. Directly, $C_0 \leq S_0$, $R_0 \leq 4S_0^2$, $M \leq 17S_0^7$, and $T_0 \leq 272S_0^{12}$, which give the claim.

6.2 Two majority reductions

Theorem 6.2 (From distribution-free samples to arithmetic circuits)**.** *Fix $d, n \geq 1$, $L \geq 3$, $\zeta > 0$, and $0 \leq \varepsilon < 1/2$. Use Theorem 6.1 with $\beta = 1/6$, and choose one $N \geq N_0$ and one $0 < \sigma^2 \leq N^{-(1+\zeta)}$, before choosing an input distribution. Suppose that for every probability distribution $\mathcal{D}$ on X_d, an iid n-sample labeled by f has positive probability of satisfying*

$$\mathcal{D}\{x : p_{S,\sigma}(f(x) \mid x) \leq 2/3\} \leq \varepsilon. \tag{3}$$

Then f has an integer arithmetic circuit with constants $-1, 0, 1$ whose sign equals f, size polynomial in $d + n + (1 - 2\varepsilon)^{-1}$, and depth $O(\log(2 + d + n + (1 - 2\varepsilon)^{-1}))$. All polynomial exponents are universal and independent of learner depth, width, scale, and ζ.

Proof. Fix $\mathcal{D}$ and choose a successful sample. Condition its posterior on $\mathcal{H}_{P_0}$, obtaining ν. On $G = \{x : p_{S,\sigma}(f(x) \mid x) > 2/3\}$, which has $\mathcal{D}$-mass at least $1 - \varepsilon$, its wrong-label probability is strictly below $(1/3)/(5/6) = 2/5$. Thus $\mathbb{E}_{h\sim\nu} f(x)h(x) > 1/5$ at every $x \in G$.

Choose an odd m_0 between $100(d + 1)$ and $100(d + 1) + 2$. Hoeffding bounds the error of m_0 independent voters at a fixed good input by $e^{-m_0/50}$. The union bound over 2^d inputs is less than one. Some fixed majority g of these low-product functions therefore agrees with f everywhere on G and has $\mathcal{D}$-error at most ε.

Let $\mathcal{K}$ be the finite class of all m_0-voter majorities of $\mathcal{H}_{P_0}$. It does not depend on $\mathcal{D}$. Writing $\kappa = 1 - 2\varepsilon > 0$, we have proved

$$\text{for every } \mathcal{D}, \qquad \max_{g\in\mathcal{K}} \mathbb{E}_{x\sim\mathcal{D}} f(x)g(x) \geq \kappa.$$

Finite minimax gives a distribution λ on $\mathcal{K}$ with $f(x)\mathbb{E}_\lambda g(x) \geq \kappa$ for every input. Choose an odd $m_1 \geq 4(d + 1)/\kappa^2$ within two of that value. Another Hoeffding bound and union bound give $2^d e^{-m_1\kappa^2/2} < 1$. Some majority of m_1 members of $\mathcal{K}$ is consequently equal to f on the entire cube.

Compile its $m_0 m_1$ low-product constituents by Theorem 1.1. Their resources are polynomial in $d + n$ by Theorem 6.1. Two further majority layers are a constant number of PNC^1 predicate-composition levels, absorbed by the closure theorem [1]. The asserted polynomial and depth bounds follow. No assertion that a majority itself has low Frobenius product has been used. □

The proof handles posterior mixtures without choosing a unique minimum-energy function. The first majority uses confidence on a large set; the second uses distribution-universal approximation and minimax. Averaging the error of individual posterior draws directly would give a weaker range of ε and would miss this distinction. Only existence of a successful sample is used, so any usual positive-probability or high-probability sample guarantee implies the hypothesis. Its distribution-free quantifier is essential.

6.3 Fixed exponents and the strong-prior limit

Corollary 6.3 (The two strong-prior regimes)**.** *The conclusion of Theorem 6.2 holds in each case below.*

(i) Fix $\gamma > 1$, put $\zeta = \gamma - 1$, choose any $N \geq N_0$, and use $\sigma^2 = N^{-\gamma}$.

(ii) Fix d, n, L, choose $N \geq N_0$ using $\zeta = 1$ and $\beta = 1/6$, and let Q_S^0 be any selected strong-prior limit of the posterior truth-table distributions as $\sigma \downarrow 0$. Assume (3) for Q_S^0 directly.

In (ii), a common subsequence for all samples may be used, and the claim includes an actual posterior limit whenever it exists.

Proof. Case (i) is immediate. For (ii), every sufficiently small σ satisfies Theorem 6.1. Truth-table space is finite and $\mathcal{H}_{P_0}$ is a fixed subset, so any limiting posterior has $Q_S^0(\mathcal{H}_{P_0}) \geq 5/6$. There are only finitely many samples of at most n inputs, so sequential compactness also permits a common subsequence. Run the proof of Theorem 6.2 on these limiting probability vectors. No passage of the strict confidence event through a limit, or independent assertion of limit uniqueness, is needed. □

The strong-prior width threshold above is sufficient; it is not claimed optimal. In both cases the chosen architecture precedes the input distribution. A proposed sample bound valid at every realizable width permits choosing this sufficiently large width after its width-independent sample budget has been fixed.

Corollary 6.4 (A conditional barrier to universal polynomial learning)**.** *Fix either a Gaussian exponent $\gamma > 1$ or the strong-prior limit of Corollary 6.3. Suppose the original learner has a distribution-free sample guarantee with some fixed $\varepsilon < 1/2$ and positive sample-success probability, bounded by one polynomial jointly in input dimension, bounded-weight teacher width and depth, and learner depth, independently of learner width, valid at every realizable learner width. Then*

$$\mathsf{P}/\mathrm{poly} = \textit{nonuniform}\ \mathsf{PNC}^1 = \mathsf{L}/\mathrm{poly}.$$

Proof. Every polynomial-size Boolean circuit family has a tanh teacher family with polynomial width and depth and uniformly bounded physical weights. Here is an explicit elementary construction. Compile gates to NAND. Carry each already computed logical bit by a neuron with activation within $1/10$ of its sign, and carry a constant $1/2$ using the weight $2\,\mathrm{atanh}(1/2)$ after its first layer. The map $\tanh(4u)$ copies a bit. A NAND of u, v is $\tanh(4(1-u-v))$, with the affine constant implemented by the charged constant wire. Its preactivation has the required sign and absolute value at least 3.2, so its output is again within $1/10$ of the desired sign. Initialize inputs by $\tanh(4x_i)$, execute one gate per layer while carrying previous bits, and multiply the final bit by two. Width and depth are polynomial in the Boolean circuit size, all weights have magnitude at most eight, and the global output margin exceeds one. Zero padding to larger widths preserves the function. If a convention requires teacher weights bounded by teacher width, pad the width to at least eight.

The hypothesized sample budget is therefore polynomial in d for every family in P/poly. Choose an admissible learner width above the ceiling threshold and apply Theorem 6.2 or Corollary 6.3. This puts the family in nonuniform $\mathsf{PNC}^1 \subseteq \mathsf{L}/\mathrm{poly}$. The reverse inclusions in P/poly are standard finite-circuit simulations, proving equality. □

This is not a known contradiction and not an exponential sample lower bound. Even a conditional failure of every polynomial bound is weaker than an $\exp(\Omega(L))$ lower bound in teacher depth. A sample guarantee polynomial in the chosen learner width, or a guarantee only for one input law, would not imply the displayed collapse.

Remark 6.5 (Other confidence thresholds). For a fixed threshold $\tau > 1/2$, choose $\beta = \tau - 1/2$ in the ceiling. Conditioning then leaves signed vote gap at least $(2\tau - 1 - \beta)/(1 - \beta) > 0$. The first majority has size $O((d+1)/\mathrm{gap}^2)$, and the same argument applies. At $\tau = 1/2$ there is no uniform positive vote gap supplied by this calculation; the same polynomial conclusion is not proved here.

7 Scope and open questions

The result isolates a tanh-specific structural restriction on the functions favored by sufficiently strong priors. Its proof is nonuniform and allows enormous original weights and widths, yet the replacement arithmetic circuit has polynomially many bits. It is not an algorithm for posterior sampling or circuit recovery.

The endpoint $\gamma = 1$ remains outside the new posterior converse. Existing descriptions of its low-action functions can use nonlinear expectation maps. A small value of that action is not a proved low-product deterministic tanh representation, so applying Theorem 1.1 to it would be an extra, unsupported step. Related circuit upper bounds in larger classes do not fill that gap.

Nor does the theorem prove that some small Boolean circuit needs exponential product or samples. It gives a conditional obstruction: universal polynomial learning would impose a strong nonuniform complexity collapse. It leaves open intermediate growth rates and does not identify an explicit hard target. A converse inclusion such as PNC^1 inside the low-product tanh class is also not established. Finally, the scalar transform extends to normalized tanh, but its Fourier formula does not automatically cover ReLU, GELU, or arbitrary Lipschitz activations.

A Width compression using ordinary tanh

We give the finite-set compression argument in full because preserving ordinary tanh, rather than merely preserving Lipschitz activations, is essential for the Fourier estimator. The proof also applies to only the first h hidden layers: replace H by h in its error and norm budgets and stop after those h steps. None of the resulting prefix-width bounds depends on the number or widths of later layers.

Theorem A.1 (Finite-set width compression for tanh)**.** *Let X be a nonempty finite set of inputs of norm at most $R \geq 1$. Suppose a depth-$D_0 \geq 2$ tanh network with linear readout, arbitrary finite hidden widths, and no later biases has product of Frobenius norms at most $P_0 \geq 1$. For every $0 < \epsilon < 1$, it has a same-depth approximation $\widetilde{z}$ such that*

$$\max_{x \in X} |z(x) - \widetilde{z}(x)| \leq \epsilon, \qquad \prod_{j=1}^{D_0} \|\widetilde{W}^j\|_F \leq 2P_0,$$

and every new hidden width is bounded by a universal polynomial in

$$P_0, D_0, R, \log(2|X|), \epsilon^{-1}.$$

The bound is independent of the original widths and individual weights. An augmented constant input is allowed and counts toward R.

Proof. Put $H = D_0 - 1$, $P_* = 2P_0$, $e = \epsilon/(8H)$ and $\eta = 1/(4H)$. We compress the hidden layers in order from the first to the last. At each step the product norm increases by at most $1 + \eta$, and the output changes by at most $6e$ on X. Thus every intermediate product is below P_*, and the final product is at most $e^{1/4}P_0 < 2P_0$, while total error is less than ϵ. A zero matrix makes the entire network zero and is trivial, so all prefix and pair norms used below may be assumed positive. If a compression step produces a zero matrix, stop and use the zero network with width one; the accumulated approximation estimates still apply.

Consider a hidden matrix W and its immediately following matrix V; the latter may be the final linear readout. Let S be the product of all Frobenius norms strictly before W, with $S = 1$ at the first layer, and put

$$A = \|W\|_F \|V\|_F.$$

The incoming feature vector $h(x)$ has norm at most SR. The part strictly after V has Lipschitz constant at most $P_*/(SA)$. If w_i^T is row i of W and v_i is column i of V, write $a_i = \|w_i\|_2$ and $b_i = \|v_i\|_2$. Zero rows or zero outgoing columns can be deleted.

Set

$$B = P_*/e, \qquad \tau^2 = 3e/(P_* R^3), \qquad \Delta = e/(P_* R), \qquad Q = (B/\tau)^2.$$

Partition neurons into large, small and middle according as $Sa_i > B$, $Sa_i < \tau$, or $\tau \le Sa_i \le B$. There are at most $S^2\|W\|_F^2/B^2$ large neurons. Their activation vector has norm at most the square root of this count. Deleting them therefore changes the final output by at most $P_*/B = e$.

For the small neurons, replace tanh by the identity. Since $|u - \tanh u| \le |u|^3/3$, their combined activation error has norm at most $\tau^2 R^3 S\|W\|_F/3$. Propagation through the suffix bounds the output error by $P_*\tau^2 R^3/3 = e$. Their resulting contribution to the next preactivation is the linear map

$$B_0 = \sum_{i \text{ small}} v_i w_i^T, \qquad \|B_0\|_* \le \sum_{i \text{ small}} a_i b_i,$$

where $\|\cdot\|_*$ is the nuclear norm.

Let $A_m = \sum_{i \text{ middle}} a_i b_i$. If $A_m < \Delta A$, delete all middle neurons: their output contribution is at most $P_* R A_m/A \le e$. Otherwise $A_m \ge \Delta A > 0$, and we approximate their contribution by sampling. Define

$$Z = \sum_{i \text{ middle}} b_i/a_i, \qquad p_i = (b_i/a_i)/Z.$$

Draw m independent indices from (p_i). A sampled neuron keeps its incoming row w_i^T and has outgoing column $v_i/(mp_i)$. If $\widehat{A} = (Z/m)\sum_{r=1}^m a_{i_r}^2$ and $c = m/Z$, its new incoming and outgoing squared Frobenius norms are exactly

$$c\widehat{A} \quad \text{and} \quad \widehat{A}/c.$$

Moreover $\mathbb{E}\widehat{A} = A_m$. Since middle row norms lie between τ/S and B/S, one has

$$Z \max_i a_i^2 \le A_m Q \le AQ.$$

The random vector $v_i \tanh(w_i^T h(x))/p_i$ has norm at most $AQSR$, uniformly on X.

Consequently some choice of the samples has both

$$\widehat{A} \le A_m + \eta A$$

and middle-preactivation approximation error at most eSA/P_* simultaneously for every $x \in X$, provided

$$m \geq 64Q^2 \max\{\eta^{-2}, (P_*R/e)^2\} \log(8(|X|+1)).$$

Here and below take integer ceilings. For detail, the scalar random variables Za_i^2/A are bounded by Q, so Hoeffding controls their mean. For a Hilbert-valued random vector bounded by $AQSR$, the mean deviation has expectation at most $AQSR/\sqrt{m}$; bounded differences gives a dimension-free tail. A union bound over X and the scalar condition has positive success probability with the displayed constant. Propagation of the vector error gives final output error at most e.

We next compress the linear branch. Truncate the singular-value decomposition of B_0 to rank

$$k = \lceil P_*R/e \rceil,$$

obtaining $B_k = UV_0$ with $\|U\|_F^2 = \|V_0\|_F^2 = \nu := \|B_k\|_* \leq \|B_0\|_*$. The residual operator norm is at most $\|B_0\|_*/(k+1) \leq A/(k+1)$, so its final output error is at most $P_*R/(k+1) \leq e$. If $\nu = 0$, omit the linear branch entirely.

When the middle branch was sampled, retain $c = m/Z$. When it was deleted and $\nu > 0$, choose $c = 1/(\nu S^2)$ instead. Represent B_k approximately using K copies of each of its at most k singular channels: their incoming rows are the rows of $\sqrt{c/K}\, V_0$, repeated K times, and their outgoing columns are the corresponding columns of $U/\sqrt{cK}$. Their total incoming and outgoing squared norms are $c\nu$ and ν/c. Their linear limit is exactly $B_k h$. The tanh linearization error at the next preactivation has norm at most

$$\frac{c\nu^2(SR)^3}{3K}.$$

After propagation this is at most $P_* c\nu S^2 R^3/(3K)$, using $\nu \leq A$. In the sampled case,

$$Z \geq A_m S^2/B^2, \qquad c\nu S^2 \leq mB^2/\Delta;$$

in the deleted case this quantity equals one. Hence it suffices to use

$$K = \left\lceil \frac{P_* R^3}{e} \max\{1, mB^2/\Delta\} \right\rceil.$$

The linear-channel approximation then contributes at most e error. When there is no sampled middle branch, the same bound is harmless; one may use the displayed deterministic value of m merely to define a common width upper bound.

Combining the sampled and linear branches gives pair product norm

$$\widehat{A} + \nu \leq A_m + \sum_{i \text{ small}} a_i b_i + \eta A \leq (1+\eta)A.$$

The last inequality is Cauchy–Schwarz over the original rows and columns. If the middle branch was deleted, the pair product is simply $\nu \leq A$. These are exact norm calculations for the constructed tanh matrices, not only bounds on their linear limits.

The compressed hidden width is at most $m + kK$. All quantities in this bound are universal polynomials in P_0, H, R, e^{-1} and $\log(2|X|)$. Crucially, no original dimension of either adjacent hidden layer occurs. Only the two adjacent matrices are altered. Processing the next hidden layer changes its own width without undoing the width bound for the preceding one. Uniform zero-padding at the end makes all hidden widths equal if desired. Summing the at most six errors and multiplying the pair-product bounds proves the theorem. □

B A logarithmic-depth ordinary-tanh normal form

We supply the geometric reduction and all remaining normalization steps. They are needed because the estimator acts on tanh itself, whereas a generic finite-precision simulation could replace tanh by lookup tables without preserving the hypothesis of the estimator.

B.1 Relevant coordinates and a two-ball cover

Lemma B.1 (Relevant-coordinate reduction)**.** *A globally unit-margin tanh representation has $P \geq 1$, and its Boolean function has at most P^2 relevant input coordinates. Fixing all the other coordinates can be done without increasing its product or reducing its global margin on the remaining cube.*

Proof. Since tanh is zero-preserving and one-Lipschitz,

$$1 \leq \mathbb{E}_x z(x)^2 \leq \left(\prod_{s=2}^{D} \|W^s\|_F^2\right) \mathbb{E}_x \left\|W^1(1,x)\right\|_2^2 = P^2,$$

where the expectation is uniform and $\mathbb{E}[(1,x)(1,x)^T] = I$. Put $T = \prod_{s=2}^{D} \|W^s\|_F$. For every relevant coordinate i, a bichromatic cube edge in that direction gives $2 \leq 2T \left\|W^1 e_i\right\|_2$. Summing squares yields at most P^2 relevant coordinates. For the irrelevant columns, choose independent uniform signs a_i, replace the first constant column w_0 by $w_0 + \sum_i a_i w_i$, and delete those columns. The expected new squared Frobenius norm equals the old squared norm. Some choice does not increase it. Every restriction has the same Boolean labels and retains the original margin. □

Lemma B.2 (Two balls)**.** *For every real matrix W, every Euclidean ball $B(a,r)$, and every coordinatewise one-Lipschitz map Φ, the set $\Phi(WB(a,r))$ is covered by two balls of radius $r\|W\|_F/\sqrt{2}$.*

Proof. Translation and scaling reduce to $a = 0, r = 1, \|W\|_F = 1$; the zero case is immediate. Let α be the leading singular value. If $\alpha^2 \leq 1/2$, one ball suffices. Otherwise put $b^2 = 1 - \alpha^2$ and $c = (2\alpha^2 - 1)/(2\alpha)$. The remaining singular part has operator norm at most b. With u a leading left singular vector, a point whose leading input coordinate is t has squared distance to the matching center cu or $-cu$ at most

$$(\alpha|t| - c)^2 + b^2(1 - t^2).$$

This is convex in $|t| \in [0,1]$ and both endpoint values equal $1/(4\alpha^2) \leq 1/2$. Apply Φ to the two centers; its Euclidean Lipschitz constant is at most one. □

For prefix products $S_j = \prod_{s \leq j} \|W^s\|_F$ and inputs of norm at most R, iteration gives at most 2^j balls of radius $RS_j 2^{-j/2}$. The remaining suffix is Lipschitz with constant at most P/S_j. Thus normalized prefix states of opposite labels are at least $2/P$ apart. If

$$j = \lceil 2\log_2(64PR) \rceil,$$

selecting one actual state from each nonempty ball gives polynomially many centers $c_a = h^j(x_a)/S_j$, each with its label f_a. Every normalized state has a same-label center within $1/(8P)$, and every opposite-label center is at least $2/P$ away. All states and centers have norm at most R. The center need not initially lie in the image: selecting an actual image doubles the cover radius, which the displayed constants allow. Every covering ball is monochromatic by the suffix Lipschitz estimate.

B.2 Physical weights and finite precision at short depth

Lemma B.3 (Bounding physical weights)**.** *Suppose a globally unit-margin tanh representation already has depth D, hidden widths at most C, and product at most P. It has a representation of the same Boolean function with no more weighted layers, product at most $2P$, margin greater than $3/2$, and individual weights bounded by a universal polynomial in d, D, C, P. It can then be rounded to dyadic weights with $O(D \log B)$ bits and product at most $4P$, where B is a polynomial bounding its dimensions, Frobenius norms, and row sums. The rounded representation retains unit margin.*

Proof. Write $M_j = \|W^j\|_F$, $S_j = \prod_{s \le j} M_s$, and $R = \sqrt{d+1}$. Margin and bounded hidden activations give $S_j \le P\sqrt{C}$ at every hidden layer. Set

$$\tau^2 = \frac{1}{16PDR^3}.$$

Replace tanh by the identity at each hidden layer whose original prefix product is less than τ. For any intermediate hybrid network, all prefix and suffix Lipschitz bounds remain valid. Using $|u - \tanh u| \le |u|^3/3$ and $\|v^{\odot 3}\|_2 \le \|v\|_2^3$, the telescoping output error is at most

$$\sum_{j:S_j<\tau} \frac{P}{S_j} \frac{(S_j R)^3}{3} \le PD\tau^2 R^3/3 \le 1/48.$$

Collapse consecutive linear blocks into the following nonlinear layer or final readout, then double the readout. If retained hidden indices are $j_1 < \cdots < j_H$, the collapsed matrices have assigned Frobenius bounds S_{j_1}, $S_{j_s}/S_{j_{s-1}}$, and $2P/S_{j_H}$. Since each retained S_{j_s} lies between τ and $P\sqrt{C}$, all weights are bounded by $\max(1, 2P\sqrt{C}/\tau)$, a polynomial. The product is at most $2P$ and the margin exceeds $3/2$. If no hidden activation remains, the affine matrix has norm at most $2P$ and the same conclusions hold.

For precision, enlarge a polynomial $B \ge 2$ to bound all dimensions, matrix Frobenius norms, and absolute row sums. At remaining depth $D' \le D$, every nonzero matrix norm satisfies

$$M_j \ge 1/(\sqrt{d+1}B^{D'-1}),$$

because $1 \le |z(x)| \le \sqrt{d+1}\prod_j M_j$. Round entries to mesh 2^{-b} with $b = O(D' \log B + \log(d+D'+2))$, with a sufficiently large universal constant. Matrix perturbations are at most $M_j/(4D')$, so the product increases by less than two. The layerwise error recursion using the row-sum bound B and Lipschitzness of tanh makes the output error less than $1/2$ at the same precision. The pre-rounding margin in fact exceeds $2(1 - 1/48) > 3/2$, so the remaining margin is greater than one. If needed, absorb the additive logarithm into $O(D \log B)$ by enlarging B. □

B.3 Replacing the long suffix by a tanh classifier

Proof of Theorem 3.3. Apply Lemma B.1, retain $k \le P^2$ inputs, and put $R = \sqrt{k+1}$ and $j = \lceil 2\log_2(256PR) \rceil$. If $D = 1$, the affine coefficients have norm at most P; double the readout and round them directly to sufficiently fine dyadic precision $O(\log(P+2))$, retaining unit margin, and finish. Hence assume $D \ge 2$. Use the prefix version of the compression theorem in Appendix A on the first $\min(j, D-1)$ hidden layers with error $1/4$, then double the readout. The global margin is at least $3/2$, the full product is at most $P_0 = 4P$, and all retained widths are bounded by a universal polynomial $C = C(P)$: the only inputs to that width bound are $P, R, j, \log(2|X_k|)$ and a fixed error tolerance. If all hidden layers have been retained, apply Lemma B.3 and finish. Otherwise the first j hidden layers have the cover and normalized centers just proved, with P_0 in place of P.

First supply a constant to a new suffix while preserving the product bound. For retained prefix products S_s, global margin gives $S_s \le P_0\sqrt{C}$. Choose

$$\epsilon_0 = 1/\max(1, 2\max_{s\le j} S_s).$$

It has inverse polynomial in P and satisfies $0 < \epsilon_0 S_s \le 1/2$. Add one constant rail whose activation in layer s is exactly $\epsilon_0 S_s$. Its first weight is $\operatorname{atanh}(\epsilon_0 S_1)$ on the augmented constant; its later weight is $\operatorname{atanh}(\epsilon_0 S_s)/(\epsilon_0 S_{s-1})$. For $0 < u \le 1/2$, $(\operatorname{atanh} u/u)^2 \le 1 + u^2 \le 5/4$. Thus the added squared norm is at most $(5/4)M_s^2$ in each layer, and the augmented prefix product is at most $(3/2)^j S_j$.

At the interface the features are $(h^j, \epsilon_0 S_j)$. A suffix written for $(u, 1) = (h^j/S_j, 1)$ is attached by multiplying its first matrix by $\operatorname{diag}(S_j^{-1} I, (\epsilon_0 S_j)^{-1})$. This increases its first Frobenius norm by at most $(\epsilon_0 S_j)^{-1}$. The complete product is therefore at most $(3/2)^j \epsilon_0^{-1}$ times the suffix's ordinary product. Both extra factors are polynomial in P.

We now give the suffix. Its input u and each of the $q = \mathrm{poly}(P)$ centers c_a have norm at most R. A correct-label center is within $1/(8P_0)$, while each opposite-label center is at distance at least $2/P_0$. In one hidden layer provide the features

$$\tanh(\beta u_i), \qquad \tanh(1 + \alpha u_i), \qquad \tanh(1 - \alpha u_i),$$

and an exact constant $1/2$. Linear combinations of these features approximate respectively u_i and u_i^2 using

$$\frac{\tanh(\beta u_i)}{\beta}, \qquad \frac{\tanh(1 + \alpha u_i) + \tanh(1 - \alpha u_i) - 2\tanh 1}{\alpha^2 \tanh'' 1}.$$

Their errors are $O(\beta^2 R^3)$ and $O(\alpha^2 R^4)$. Choose α, β inverse-polynomially small in C, R, P_0 so that every affine combination

$$D_a(u) = \sum_i \mathrm{approx}(u_i^2) - 2\sum_i (c_a)_i \,\mathrm{approx}(u_i) + \|c_a\|_2^2$$

approximates $\|u - c_a\|_2^2$ to error at most $1/(256P_0^2)$. This is possible with polynomial weights because $\|c_a\|_1 \le \sqrt{C}R$ and $\tanh'' 1 \ne 0$. Every affine constant is charged to the constant rail.

In a second hidden layer use

$$t_a(u) = \tanh(B_0[\tau_0 - D_a(u)]), \qquad \tau_0 = 1/(16P_0^2), \quad B_0 = 32P_0^2 \log(16q),$$

and propagate the constant $1/2$. Put $\chi_a = (1 + t_a)/2$. For a selected close center, $\tau_0 - D_a \ge 11/(256P_0^2)$; for every opposite-label center it is less than $-3/P_0^2$. The tanh tail estimate gives $\chi_a \ge 1 - 1/(16q)$ for a close center and $\chi_a \le 1/(16q)$ for each opposite-label center. All other χ_a lie in $[0, 1]$, so

$$f(u)\sum_a f_a \chi_a(u) \ge 1 - 1/(16q) - q/(16q) \ge 7/8.$$

The readout $2\sum_a f_a\chi_a = \sum_a f_a t_a + \sum_a f_a$ has margin at least $7/4$; implement its constant by the carried $1/2$ rail. This is an ordinary-tanh network with two hidden layers, polynomial width, polynomial weights, and polynomial product.

Attach it to the augmented prefix. The resulting network has depth $j + 3 = O(\log(P + 2))$ and polynomial width and product. Some interface weights can be large if S_j is tiny. Apply Lemma B.3 to this now-short network. It preserves ordinary tanh and the Boolean function, makes all physical weights polynomial in P, and rounds them with $O(D' \log B) = O(\log^2(P + 2))$ bits, since both the new depth D' and $\log B$ are $O(\log(P + 2))$. This proves all parts of the normal form. □

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